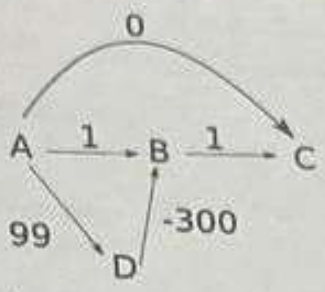
1. Bellman-ford algorithm can be used to specify the shortest path in the graph below and it will needs _____ passes.



2. If $f(n) = \log n$, then $f(n) = \omega$ (_____)

3. _____algorithm can be used to identify the strongly connected components.

4. The number of edges in a spanning tree of n vertex undirected graph is _____ edges.

5. _____ is a stable and in-place sorting algorithm.

6. Consider the recurrence: $T(n) = T(n/4) + 2n$, the size of each subproblem generated at each step is _____ the size of the original problem.

7. The asymptotic relationship between $T_1(n) = 4T(n/2) + n^2$ and $T_2(n) = T(n/2) + 1$ is $T_1(n) =$ _____ $(T_2(n))$.

8. To apply _____ sort algorithm, each input should be an integer.

9. If $T(n) = 2\,T(n/2) + n$, then case _____ of master method will be applied.

10. The MAX-HEAPIFY procedure in max-heap runs in $O($_____).

11. If $f(n) = o(g(n))$, then _____ grows faster.

12. The optimal solutions for searching on unsorted data by comparing keys require $\theta($_ comparisons.

13. If $f(n) = n^4 + n$ and $g(n) = 2^n + n$, then $f(n)+g(n) = O($ _____)

14. _____ is a sorting algorithm that follows the divide and conquer approac

A. Choose one correct answer for each statement and fill the following table: [3 marks]

Question	1	2	3	4
Answer				

1. **If a problem R is NP-Complete, then:**
   a) R is NP
   b) R is NP-hard
   c) Both answers **a** and **b** are correct.
   d) Both answers **a** and **b** are incorrect.

2. **Choose the correct statement:**
   a) Decision problems can be maximization or minimization problems.
   b) If any NP-complete problem can be solved in polynomial time, then every problem in NP can also be solved in polynomial time.
   c) Heuristics may produce good solutions and guarantee the quality of their solution.
   d) It has been proven that P=NP.

3. **The class P consists of:**
   a) all those problems that are solvable in polynomial time.
   b) problems that can be solved in time $O(k^n)$, where n is the size of the input to the problem.
   c) finding a Hamiltonian cycle problem.
   d) All the above answers are correct.

4. **Suppose $A \leq_p B$. Which of the following statements is not correct?**
   a) If B is P, then A is P
   b) If A is NP, then B is NP.
   c) If A is NP-hard then B is NP-hard
   d) If A is NP-complete, then B is NP-complete.

B. Circle the correct answer in each of the following statements: [2 marks]

A. The graph coloring problem is **[tractable / intractable]** problem.

B. To solve the 3-SAT problem in reasonable time, you should use an **[approximation algorithm / exact polynomial-time algorithm]**.

C. Finding the **[shortest / longest]** path is an NP-complete problem.

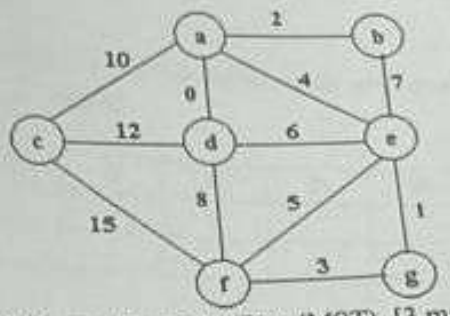
D. To prove that TSP is an NP-hard problem, we need to show that **[TSP $\leq_p$ HAM-CYCLE / HAM-CYCLE $\leq_p$ TSP]**.

To be answered in (30) Minutes

A. **Given the following array {127, 324, 173, 4,38, 217, 134}. Apply the task is to sort them using Radix sort. Show your work in the table below given each pass.** [2.5 marks]

Array	127	324	173	4	38	217	134
Pass 1							
Pass 2							
Pass 3							

B. **Use undirected-weighted graph below to find a minimum spanning tree (MST) by using Kruskal algorithm.**



1. Draw the resultant Minimum Spanning Tree (MST). [3 marks]

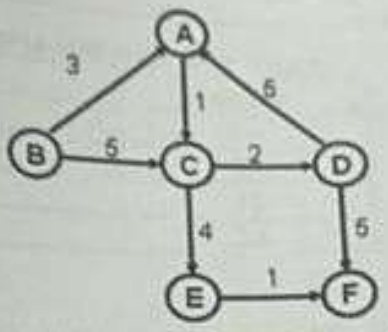
2. Specify the order in which edges get added to the spanning tree? [1 mark]

3. Specify the value of MCST. _____ [0.5 mark]

4. If Prim's algorithm were to be applied on the graph above, which will be the 5th edge to be added onto the spanning tree being formed. Start from the vertex 'c'? _____ [1 mark]

Student ID:

C. **Apply Dijkstra's algorithm to find the shortest distance between vertex (B) and all other vertices.**



1. What is the order in which vertices get removed from the priority queue? [1 mark]

2. What is the resulting shortest-path tree including the total distance from B to each another vertex? Fill the table. [3 marks]

Vertex	A	B	C	D	E	F
key						
π						

Problem

Student ID: _____

To be answered in (30) Minutes [ ] / 10 Marks

Consider the following knapsack problem with four items:

item	weight	benefit
1	2	16
2	3	15
3	4	12
4	6	24

The capacity of the knapsack is 8.

A. If the above knapsack is <u>0/1 knapsack</u>, solve the problem using dynamic programming.

1. Fill the table below. [4 marks]

	0	1	2	3	4	5	6	7	8
0									
1									
2									
3									
4									

2. Which items are included in the knapsack? Show on the above matrix the steps you have followed. [1 mark]

3. What is the total profit? _____ [1 mark]

B. If the above problem is a <u>*fractional knapsack*</u> problem, answer the following questions.

1. Which items are included in the knapsack? Show the details. [3 marks]

2. What is the total benefit of the items included in the knapsack? _____ [1 mark]